

One local, grounded AI assistant brings product Q&A, recognition, support and planning into a single experience

OUR APPROACH



Local-first and private

Questions, images and audio are processed on the user's machine by open models. No cloud AI service and no API key.



Grounded, not guessed

Answers come from retrieved KOHLER support documents and show which sources were used.



Multimodal by design

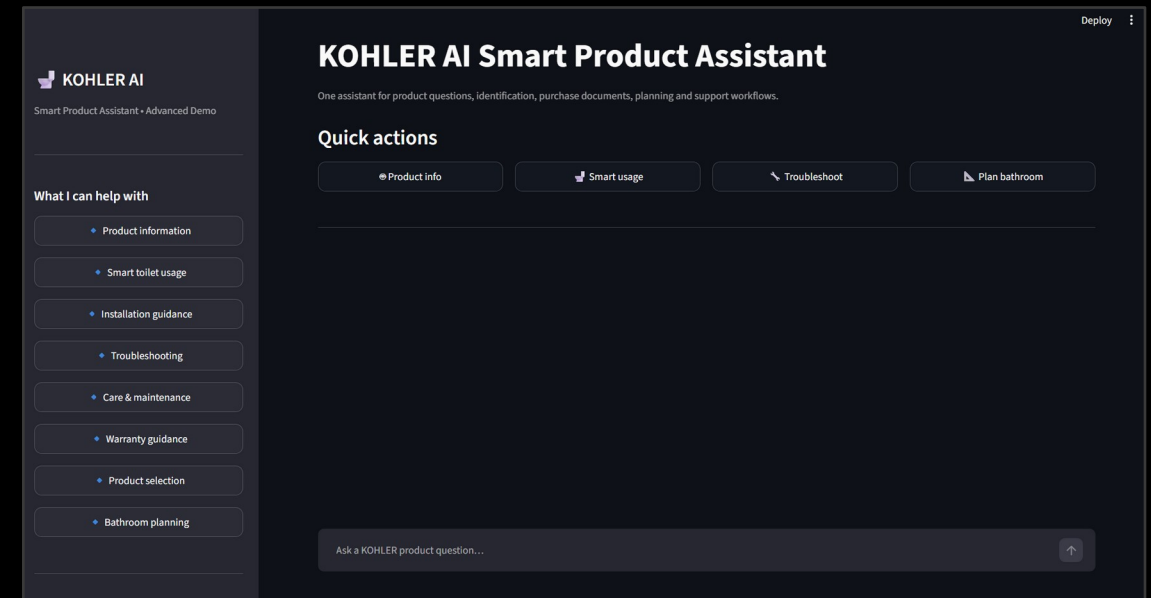
One vision-language model handles text questions, product photos and invoice images.



Assistive and safe

Rule-based first checks for faults, and every result prompts verification against the label or official KOHLER support.

THE PRODUCT · LIVE UI FROM THE REPO



ONE ASSISTANT · NINE CAPABILITIES · FOUR JOBS TO BE DONE



ASK

AI chat with intent detection
Voice assistant



SEE

Product scanner (photo to model)
Bill scanner (invoice to JSON)



FIX

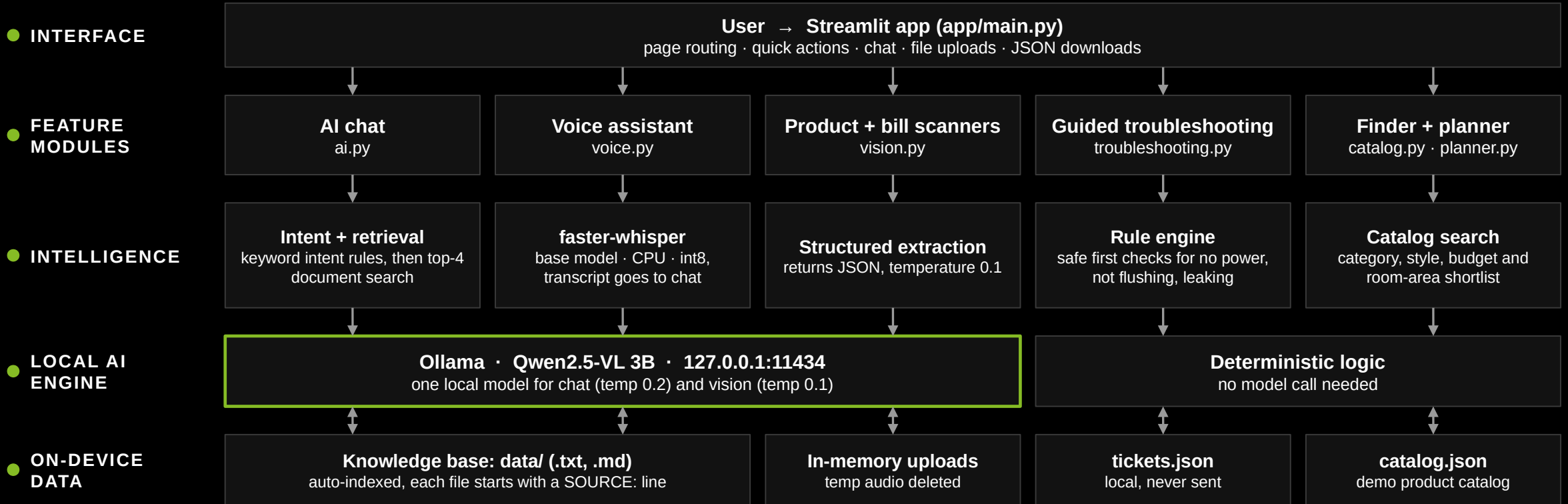
Guided troubleshooting + ticket
Manual and support finder



CHOOSE

Product finder by style, budget
Bathroom planner

A layered architecture keeps retrieval, reasoning and data entirely on the user's own machine



CHAT PIPELINE IN FOUR STEPS

- 1 Detect intent**
usage, troubleshooting, warranty, installation, care or purchase
- 2 Retrieve**
TF-IDF cosine blended with keyword overlap, top 4 docs kept
- 3 Generate**
local model gets docs, last 8 messages and a no-invention prompt
- 4 Answer**
shown with the detected intent and the sources used

A lean open-source stack delivers multimodal AI with no cloud service and no API key

LAYER	TECHNOLOGY	ROLE IN THE SYSTEM
Interface	Streamlit	Web UI, page routing, file uploads and JSON downloads
Language	Python 3.10+	Single-language codebase with 12 focused modules under app/
Local AI runtime	Ollama	Serves the model over a local API on 127.0.0.1:11434
Vision-language model	Qwen2.5-VL 3B	One model for chat, product photos and invoice extraction
Speech to text	faster-whisper	Base model on CPU (int8) transcribes wav, mp3 and m4a locally
Retrieval	TF-IDF, scikit-learn	Unigram and bigram cosine similarity blended with keyword overlap
Configuration	python-dotenv	Model, URL and timeout set through .env (180 s default)
Quality	pytest	Tests for retrieval and intent detection

0

API keys or cloud AI calls

3.2 GB

one-time local model download

Top 4

documents retrieved per question

8

recent messages kept as context

LOCAL VISION OUTPUT · PRODUCT SCANNER

```
Identification result
{
  "product_type": "Smart Toilet"
  "likely_model": "K-5000"
  "visible_brand": "KOHLER"
  "visible_model_number": "K-5000"
  "visible_features": [
    0: "Touchless Lid"
    1: "Remote Control"
    2: "LED Display"
    3: "Hygienic Seat"
  ]
  "confidence": 0.95
  "next_step": "No additional photo/label is needed."
}
```

Local-first multimodal support AI is the differentiator, with a clear path from prototype to production



Zero-cloud multimodal AI

Chat, vision and speech all run on-device with open models. No API key, no cloud AI service, no data leaving the machine.



Guardrails built into the prompt

The model may not invent models, prices, warranty eligibility or repair steps. Retrieved text is untrusted, and every answer cites its sources.



Photo-to-model and bill-to-JSON

One vision model identifies a product from a photo and turns an invoice into structured JSON, ready to download.



Diagnose-to-ticket loop

Guided troubleshooting runs safe first checks, then saves a support ticket locally as JSON for easy handoff.

Extract bill details

Extracted information

```
{
  "merchant": "KOHLER BATHROOM SOLUTIONS",
  "invoice_number": "KHL-DEMO-2026-0919",
  "invoice_date": "19 September 2026",
  "customer_name": "Demo Customer",
  "products": [
    {
      "description": "KOHLER Smart Toilet",
      "model_number": "DEMO-KOHLER-SMART-01",
      "quantity": "1",
      "amount": "145,000"
    },
    {
      "description": "Professional Installation",
      "model_number": "INST-STD-01",
      "quantity": "1",
      "amount": "8,500"
    }
  ]
}
```

Bill scanner: synthetic demo invoice turned into structured JSON

FROM PROTOTYPE TO PRODUCTION (ROADMAP FROM THE REPO)

● TODAY · Prototype

Local RAG chat, product and bill scanners, voice, troubleshooting, finder and planner, all working in the demo.

● NEXT · Data depth

Real-time KOHLER catalog and pricing, stronger invoice OCR, and PDF manual ingestion into the knowledge base.

● LATER · Service integration

Clearance checks and visual layout, service-centre and CRM integration, and regional (India) product and warranty data.